

Cloudflare Radar

2024 Year in Review

Global Internet Traffic, Adoption Trends, Connectivity & Security Insights

Prepared for Technology Executives & Network Infrastructure Leaders

17% Global Traffic Growth and Starlink's 3.3x Surge

17%

Year-Over-Year Traffic Growth

Global Internet traffic continued its upward trajectory in 2024, driven by increasing reliance on the Internet for communication, commerce, entertainment, and transportation.

3.3x

Starlink Traffic Surge

SpaceX Starlink brought satellite Internet connectivity to previously underserved locations, with traffic growing 3.3x during 2024.

Key Growth Drivers:

- Mobile devices accounted for 41% of global traffic
- Spain recorded the highest download speeds among measured countries
- IPv6 adoption reached 28% of dual-stack traffic
- Increasing reliance on Internet for commerce, entertainment & transportation

Google Leads Top 10 Services as Generative AI Rises

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Generative AI Continues Rapid Rise

Generative AI saw meteoric growth throughout 2024, continuing its trajectory as one of the most transformative Internet trends.

Google maintained its position as the #1 most popular Internet service for the second consecutive year, based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver.

While the Metaverse receded into the background, AI-powered services drove significant increases in traffic and user engagement.

Search Engine Market Share:

- Google: 89% of referral traffic
- Yandex: 3.1% • Baidu: 2.7% • Bing: 2.6%
- DuckDuckGo: 0.95%

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

2% → 13%

Post-Quantum Encryption

TLS 1.3 traffic now uses post-quantum encryption following Chrome 124 default enablement (April 17, 2024).

21%

HTTP/3 Adoption

HTTP/3 runs on QUIC, providing faster connections, packet-loss mitigation, and encryption by default.

12%

Go API Clients

Go surpassed Node.js and Python to become the #1 API client language with ~12% of automated API traffic.

Additional Adoption Metrics:

- Mobile OS: iOS devices account for 33% of traffic; Android exceeds 95% peak share in some regions
- Post-quantum rollout: Chrome 124 enabled by default → rapid climb from 2% to 13%; Firefox & Safari following
- Go surpasses Node.js, Python (9.6%), Java (7.4%), and .NET (3.6%) in API client usage
- Chrome remains the most popular browser globally

225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

Major Internet Disruptions in 2024

Cause	Count
Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1

Key Insight: Government-Directed Shutdowns

Government-directed shutdowns emerged as the leading cause of major Internet disruptions in 2024, accounting for more than half of all recorded incidents.

This underscores the increasing use of Internet shutdowns as a tool for controlling information flow during political events, protests, or national security situations.

The scale of disruptions — 116+ government-directed — reflects a significant challenge for global connectivity and digital economy.

Secondary Factors:

- Cable cuts (41 incidents) highlight infrastructure vulnerability
- Power outages (16) show energy dependency
- Combined, non-government causes represent 109 disruptions

Cloudflare's Radar Outage Center tracks scope and duration of these disruptions using real-time traffic data.

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

Traffic Mitigated

68%

Bot Traffic from Top 10 Countries

4.3%

Malicious Emails

43%

Deceptive Links in Malicious Emails

Most Attacked Sectors & Threats

- Gambling/Games sector was the most attacked industry category
- Log4j (CVE-2021-44228) exploitation persists 2+ years after disclosure
- RPKI valid routes increased by 6.4%, improving routing security
- .bar TLD originated the largest share of malicious & spam email
- 43% of malicious emails contained deceptive links to trick recipients

Bot Traffic Concentration

68% of global bot traffic originated from just the top 10 countries, highlighting concentrated threat sources.

Bot traffic represents a significant portion of automated attacks, credential stuffing, and scraping operations across the Internet.

Understanding bot concentration is critical for threat intelligence and defensive strategy.

Five Takeaways Shaping the 2025 Internet Outlook

1

Traffic Growth Trajectory

Global Internet traffic grew 17% YoY, with Starlink surging 3.3x as satellite connectivity expands to underserved regions.

2

Post-Quantum Encryption Momentum

Post-quantum encryption adoption leapt from ~2% to 13% of TLS 1.3 traffic following Chrome 124's default enablement in April 2024.

3

Connectivity Fragility

Government-directed shutdowns remained the leading cause of the 225 major Internet disruptions observed, followed by cable cuts (41) and power outages (16).

4

Persistent Security Threats

6.5% of all traffic was mitigated; Gambling/Games was the most targeted sector; Log4j exploitation persists years after disclosure.

5

Evolving Developer Ecosystem

Go surpassed other languages to become the most popular API client language, capturing ~12% of automated API traffic.